SECTION III—LOCOMOTIVE LUBRICATION AND TYPES OF LUBRICATORS

Efficient locomotive operation requires the proper lubrication of the various parts in order to dissipate the effect of friction or that force which acts between two bodies at their surfaces of contact to resist their sliding upon each other. Lubricants of various kinds are used to reduce the resistance and heating, caused by motion of the parts in contact. The character of the lubricant and the method of application depend upon the conditions to be met.

With the use of superheated steam, difficulty was experienced because of unsuitable piston rod and valve stem packing, and the carbonization of cylinder lubricants. The packing has been improved and high-grade lubricants have been compounded to withstand the increased temperature conditions.

It is more important that the oil be properly applied to the valves and cylinders than that the oil be of a particular type.

The value of a lubricant to decrease friction depends upon its ability to maintain a film sufficient to prevent the surfaces from coming in contact with each other.

Light oils give best results where the bearing pressures are low and speeds high. Oils of this type are obtained almost entirely from the mineral products, although some are of a vegetable origin. Heavy grades of oil are necessary under conditions where the load and speed would crush and wear out the lighter oils.

Greases and solid lubricants, such as animal and vegetable fats, soapstone, graphite, etc., are used for slow moving bearings under heavy loads.

Grease, in the form of compressed cakes, has been generally used for lubricating driving box bearings. Hard grease has also been used for main and side rod pin lubrication. Soft grease, injected by a lubricating compressor or a compressed air gun, is now used for rods and many other bearings on a locomotive.

Satisfactory lubrication depends as much upon the method of applying the lubricant as upon the use of a suitable lubricant. Various methods of application are and have been in satisfactory use in locomotive service meeting the particular needs of design and operating conditions. Gravity, force, splash and capillary action feeds have been used to apply the oils, while friction and heat have been employed in the application of greases.

Formerly the locomotive cylinders and steam valves were lubricated solely by means of the hydrostatic lubrication. However, the mechanical type of lubricator has for some time been employed by many roads on locomotives for any kind of service.

Formerly a separate lubricator was used for lubricating the steam cylinders of the air compressor, but in modern practice this lubricator is incorporated with the main lubricator which supplies oil to the locomotive cylinders and valves.

Mechanical lubricators are also used for supplying oil to the guides, driving box bearings, and the shoes and wedges.

Flange lubricators have been developed to meet the conditions of locomotives with a long wheel base and where the roads have considerable track curvature.

Hydrostatic Lubricators

The hydrostatic system of lubrication, for oiling the valves and cylinders of locomotives, is based upon the principle of hydrostatic displacement, utilizing the pressure due to a head of water, together with the difference in the specific gravities of oil and water, to force the oil from the oil reservoir into the feed pipes.

When this principle was applied to locomotives for the oiling of valves and cylinders, convenience and accessibility made it desirable to take the steam supply directly from the boiler and connect the oil pipes to the top of the steam chest. Connecting the lubricator in this manner resulted in full boiler pressure prevailing at all times at the inlet end of the lubricator, while at the outlet end the pressure was variable. This variation in pressure depended on whether the locomotive was running under steam or with the thottle closed as when drifting.

To overcome this variation in pressure at the outlet end of the lubricator, an equalizing feature was devised. This feature consists of a steam chamber located near the delivery end of the lubricator. The chamber is at all times subjected to direct boiler pressure, from which the steam and oil are permitted to escape through a passage of

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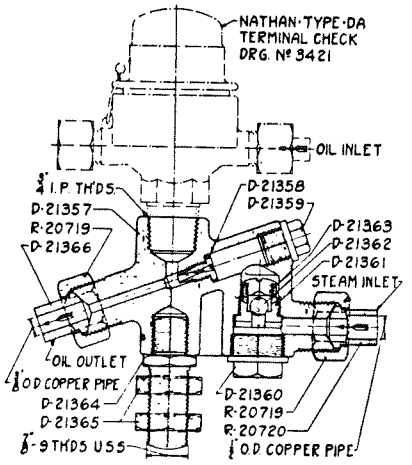
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atomizing nozzle to remove any dirt or lime deposit. No choke is used in the oil line at the delivery point.

The preferred method of application is for two atomizers to be located on top of each steam chest, one for the valve and one for the cylinder. They may be supported on the same brackets used for the



NAMES OF PARTS

Ref. No.	Part
D-21357	Body
D-21358	Nozzle
D-21359	Nozzle Cap
D-21360	Check Cap
D-21361	Check Retainer
D-21362	Ball
D-21363	Stop
D-21364	Mounting Bolt
D-21365	Nut
D-21366	Tailpiece (⅜" C. P.)
R-20719	Coupling Nut
R-20720	Tailpiece (½" C. P.)

Fig. 43. Nathan Type 3477 Oil Atomizer

terminal checks. The terminal checks are screwed directly into the atomizers. Saturated steam should be used for the atomizer, preferably from the air pump steam line, when saturated steam is used for the pump. Live steam from a separate source, such as the lubricator heater line, may also be used. The steam supply must remain open while the locomotive is on the road.

Maintenance of the Nathan DV Mechanical Lubricator. It is recommended that the lubricator be filled to the top mark on the gauge glass frame before each trip. The frame of the gauge glass at the front of the lubricator is graduated in pints, so that the amount of oil fed out on each trip can be easily ascertained. As a general rule, it is recommended that lubricators be filled only to the highest visible point of the gauge glass. This will provide adequate space for expansion when the oil is heated. In severe cold weather it is advisable to warm the oil before pouring it into the lubricator.

Clean around the filling plug, before removing, to prevent cinders and foreign matter from getting into the lubricator. Do not open the cover for filling; always use the filling plug. Only clean oil should be used in the lubricator.

The presence of water in the oil chamber may be easily ascertained by means of the waste cock *80,* Fig. 29. On the DV 5, 6, and 7 sizes, a body plug is also located at the bottom of the heating chamber for completely draining the heater compartment. Water in the lubricator is harmful, therefore the drain should be opened occasionally to allow all water to drain out.

If the report on the lubricator indicates water in the body, lack of lubrication, leaky pipes or any mechanical defects, the lubricator should be checked and repaired in the following manner:

A leaky terminal check is indicated by the temperature of the oil pipes as stated previously. Water in the lubricator may indicate a leaky terminal check. A test gauge should be applied, as shown in Fig. 44, using the connection on the terminal check provided for this purpose. When testing and setting terminal checks on the locomotive, the oil in the lubricator should be at a temperature of not less than 100° F. When tested on a test rack, a light oil should be used for this purpose. If pressure cannot be maintained, the terminal check should be replaced or fitted with a new pin valve and seat. In assembling the check, after inspection and repair, the diaphragm plates should be cleaned thoroughly, and must form a perfectly tight joint between the body and the diaphragm ring. This is to prevent oil from leaking to the top side of the diaphragm.

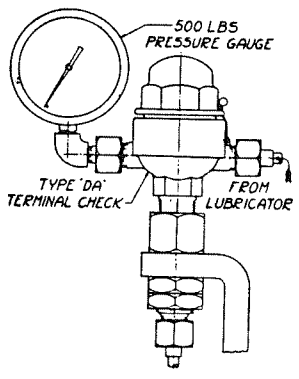


Fig. 44. Method of Applying Test Gauge to Nathan Terminal Check

ing the lubricator camshaft, if the gauge pressure drops rapidly it indicates that the plunger cylinder cup needs replacement. If considerable hand cranking is necessary to register on gauge pressure, the plunger cylinder cup may need replacement, or perhaps the cylinder strainer *L-591-2*, Fig. 86, area is reduced by accumulated dirt, lint, and cinders in the bottom of the lubricator body, in which case wash out the body with kerosene.

To definitely determine a lubricator's performance, it should be tested and proved on a shop test rack.

Fig. 88. Shows quantity delivered for one revolution of camshaft with various adjustments of feed adjustment stem for one lubricator pump.

Fig. 89. Shows oil quantity delivery in pints per 100 miles for one lubricator pump feed.

Nathan Steam-Operated Lubricators for Feed-Water Heaters

Fig. 90 shows the Nathan Type LB-2A steam-operated lubricator designed to lubricate the feed-water pump steam cylinders.

Fig. 91 shows the application of this lubricator to the Worthington Type SA Feed-Water Heater.

The amount of oil delivered by the lubricator is controlled by the speed of the feed-water pump, and the oil-feed starts and stops with the feed pump.

The lubricator is a self-contained device attached to the side of the steam pump. It derives its motion from the pulsations of the steam pressure in the pump cylinder. The movement of the small steam piston *55*, Fig. 90, is controlled by the action of levers *11* and *15* and return lever spring *117* and actuates a ratchet mechanism and a crosshead *8*, which imparts a reciprocating and oscillating motion to a pumping unit.

The pumping unit is valveless, similar to the units used in the Nathan DV Lubricator previously described. The discharge from the pumping unit is piped to a terminal check located in the feed pump steam line. The amount of oil fed is adjusted by the regulating screw 30. By turning this screw to the left the oil feed is increased and by turning it to the right it is decreased. The stroke *H* is equal to the distance between the regulating screw *30* and the lock nut *32*, therefore the adjustment may be readily checked. The stroke of the pumping unit when wide open is $\frac{5}{16}$".

A crank handle *7* is provided for filling the oil pipes and also indicates the movement of the lubricator.

The steam piston *55* is provided with a deflector *59* to prevent condensate from entering the oil reservoir. The condensate is discharged through the drain elbow *61*. The lubricator should be filled with valve oil through the strainer *38*. Fill only to the top of the strainer to permit expansion when the oil is heated. Steam connections are provided for heating the lubricator in cold climates.

Nathan Air Compressor Lubricators

Figs. 92 and 93 show the Nathan Type LB Air Compressor Lubricators, and Figs. 93 and 95 illustrate their application to the air compressors.

In the class LB-2, two pumping units *P*, Fig. 92, are provided, one for the steam cylinders, the other for the air cylinders. These pumping units are again similar to the units of the DV lubricator. The pistons *T* combine a reciprocating and oscillating movement whereby the intake and discharge ports are automatically opened and closed.

The lubricator is operated by air pressure from the air pump low-pressure cylinder. The piston *K* is reciprocated by the changing pressure from the air cylinder and the return spring *F*. This movement is transmitted to the pump pistons *T* by means of the pawl *Z*, the driving shaft *W*, and the crosshead *U*. The lubricator makes one stroke while the air compressor makes 18 double strokes.

The regulating screws *R-1* and *R-2* serve for adjusting the oil feed.

The stroke "h" should be normally $\frac{1}{32}$" for the outlet *1* (lubrication of the air cylinders) and at $\frac{5}{16}$" for the outlet *2* (lubrication of the steam cylinders). The regulating screw for the feed to the air cylinders is correctly set at the factory and the feed should not be increased. The regulating screw for the feed to the steam cylinders may be reduced as required.

The LB-3 lubricator has four pumping units, two for the steam end and two for the air end of the compressor. It is provided with two separate reservoirs to permit using different oils for the air and steam ends of the compressor. The discharge from one of the pumping units for the steam feeds is piped to a terminal check located in the air pump steam line, the second steam feed, when used, is con-